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FACULTY OF ENGINEERING
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CSE4088 Introduction to Machine Learning
Term Project - Model Development Report

Project Title

Email Spam Detection

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1. Overview

This report provides an overview of the progress made in developing machine learning models for detecting spam messages using the preprocessed dataset. The task involves loading the dataset, specifying features and labels, building and training multiple models, optimizing their performance, and saving the results for the next step, model evaluation.

2. Work Completed

2.1. Data Preprocessing

The dataset was cleaned by handling missing values, normalizing features, and encoding categorical variables.

The data was then split into training and test sets to ensure proper model validation.

2.2. Data Splitting

The dataset was split into training and test sets with 80% training and 20% testing using `train_test_split`.

2.3. Model Definition and Training

Four machine learning models were defined for spam classification. The first model is a Random Forest Classifier, consisting of 300 decision trees and using a random state of 42. The second model is Logistic Regression, which uses the default solver and a random state of 42. The third model is Multinomial Naive Bayes, it is well suited for text classification, particularly when dealing with word frequency features. Lastly, the K-Nearest Neighbors model was built, with 5 neighbors. Each model offers different strengths for the spam classification task.

2.4. Prediction

Using the test dataset for all four models, predictions were made, and the prediction times were recorded. The performance of each model was evaluated using accuracy and classification reports.

2.5. Model Evaluation

Accuracy scores and detailed classification reports, including precision, recall and F1-score, were generated for each model to evaluate their effectiveness in classifying spam messages.

2.6. Model Optimization

Hyperparameter tuning was performed for each model using GridSearchCV and RandomizedSearchCV. For each model, a parameter grid was defined to explore hyperparameters in order to specify the best combination of values that yields the best performance.

2.7. Saving Models

After model evaluation and model optimization, all models were saved using “joblib”, including the best-performing ones. Additionally, the prediction results were saved as CSV files.

2.8. Prediction Results

The prediction results for all four models were saved for further analysis, allowing for a comparison between these models.

3. Challenges Faced

3.1. Issues with the Dataset

- **Class Imbalance:** The dataset contained significantly more non-spam (ham) messages than spam messages, which created challenges for model training. This imbalance affected the recall for the spam class and resulted in misclassifications.
- **Noisy Data:** Some messages contained typos, unusual formatting, or non-standard characters, which complicated feature extraction and preprocessing.
- **Limited Features:** Although engineered features like message length and keyword counts were useful, additional contextual features (e.g., sender

metadata or sentiment analysis) could have further improved classification accuracy.

3.2. Computational Limitations

- **Optimization Time:** GridSearchCV and RandomizedSearchCV required substantial computational time for hyperparameter tuning, particularly for models like Random Forest.
- **Memory Usage:** k-Nearest Neighbors consumed significant memory when working with high-dimensional data, leading to slower evaluation times.

3.3. Model-Specific Challenges

- **Overfitting:** Random Forest, with a high number of trees, sometimes overfitted to the training data, requiring careful tuning of hyperparameters.
- **Low Recall in Logistic Regression:** The linear nature of Logistic Regression struggled to capture non-linear relationships in the data, reducing recall for the spam class.

3.4. Prediction and Evaluation

- **Misclassification of Subtle Spam:** Models occasionally failed to identify subtle spam patterns, such as legitimate-looking emails containing harmful links. This highlighted the need for more advanced features or external datasets to enrich model training.